Workshop Registration and Notifications

FUNCTIONALITY DOCUMENT | Backend Task 2

1. Purpose

Build a workshop registration service and a mock HTTP notification service. A successful registration must eventually produce a confirmation, and cancelling a workshop must notify all registered members.

2. Functional Requirements

- Allow organizers to create workshops and members to register.
- Allow each member to register only once per workshop.
- Return the registration result without waiting for the notification service.
- Ensure every successful registration eventually produces a confirmation; failed registrations must not produce confirmations.
- If the backend restarts after saving a registration, its confirmation must still be sent.
- Retry notification delivery when the mock service is unavailable.
- Ensure a notification failure for one member does not block delivery to other members.
- When a workshop is cancelled, notify every registered member.
- Provide a mock notification service that sometimes returns an error or times out. A successful response represents successful delivery.

3. Required Backend Behaviour

Registration

Create workshops and accept member registrations while preventing duplicate registration by the same member for the same workshop. Return the registration result independently of notification delivery.

Notification delivery

Each successful registration must eventually trigger a confirmation. Failed registrations must not trigger confirmations. Delivery must be retried when the mock notification service is unavailable.

Persistence and recovery

A pending confirmation must remain deliverable if the backend restarts after the registration has been saved.

Cancellation

Cancelling a workshop must trigger notifications to all registered members.

Failure isolation

A failure delivering a notification to one member must not prevent attempts to deliver notifications to others.

4. Mock Notification Service

Implement a mock HTTP service that can sometimes return an error or time out. Treat a successful response from this service as successful delivery. No real email or SMS integration is required.

5. Required Demonstration and Submission

- Demonstrate registration and workshop cancellation.
- Demonstrate notification failure and recovery.
- Submit source code and clear instructions for running the services.
- Explain how the design handles lost notifications, retries, duplicate delivery, and backend restarts.

6. Scope Boundary

This document lists only the functionality specified in the GDG TECH Wildcard Entry brief for Backend Task 2. Real email/SMS integration and a separate frontend are not required by the brief.